

R E V I E W

The role of dermocosmetics in the management of rosacea: An up-to-date

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ABSTRACT

Dermocosmetic interventions represent an effective support to pharmacology therapy of rosacea, in preventing disease flare-ups, enhancing skin barrier function, and helping reduce inflammation during acute phases. The aim of this article is to report on the current scientific consensus on the role of dermocosmetics in the management of this disease. A thorough PubMed literature review on the use of dermocosmetics in this disease highlighted two key aspects: the lack of patient awareness regarding the natural history of their condition and the significant gap in patient and physician education concerning the role of dermocosmetics and skincare as strategies in the management of rosacea. Effective skincare is mandatory both in the prodromal and the flare-up phases of the disease. Daily care is structured in morning and evening procedures aiming to improve barrier dysfunction, to restore skin hydration, to normalize skin pH, to improve the symptoms of rosacea and patient's social interaction. The suggested protocols are based on the use of appropriate cleansers, moisturizers, antioxidant serum, adequate photo-protection and camouflage.

Key words: Rosacea, Pathogenesis, Skincare, Dermocosmetics, Treatment, Management



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Introduction

Rosacea is a common chronic inflammatory skin disease with a prevalence of 5.46% of adults worldwide, but in consideration of the high number of paucisymptomatic patients this figure is realistically greater.

Frequent relapses characterize the natural history of the disease, which remains treatable but not yet curable¹. Since the most recent updates from the American Acne and Rosacea Society (AARS)² and the Global Rosacea Consensus (ROSCO)³, the 2020 *Beyond the Visible* report has further emphasized the significant psychological and non-visible burden associated with rosacea⁴.

In light of current evidence, a multidisciplinary and integrated approach appears crucial for optimizing rosacea management and addressing the multifaceted needs of affected patients⁵⁻⁷. Patient education is a foundational element of care, facilitating awareness of the condition and reinforcing the importance of lifestyle modifications and a personalized daily skincare regimen as essential components of a comprehensive therapeutic strategy.

In practice, the patient is often not aware of the importance of combining an appropriate skincare with the pharmacological treatment. In this regard, Aesthetic Medicine expertise plays an important role in guiding patients toward appropriate use of both dermocosmetics and cosmeceuticals. In practice, these two categories of products are not always easily distinguishable from each other within the broad cosmetics market. In the treatment of rosacea, the goal of cosmetics is to allow normal skin hygiene and improve the patient's self-image without damaging, while the goal of cosmeceuticals is to reduce subjective symptoms, such as pruritus, burning, tightening of the skin⁸⁻¹⁰.

The aim of our work is to review the scientific literature published over the last 10 years to evaluate the approach of the scientific community on the role of dermocosmetics in the management of rosacea and to propose to the Aesthetic Physician a shared dermocosmetic protocol based on current evidence.

Materials and methods

Our work was based on a thorough literature review, selecting scientific articles published between 2015 and

2024. The primary databases we referred to were PMC (PubMed Central), NIH (National Library of Medicine) and ClinicalTrials.gov. The articles were selected using the following keywords: "rosacea", "pathogenesis", "skincare", "dermocosmetics", "treatment" and "management".

The inclusion criteria encompassed original research articles, systematic and narrative reviews, and clinical trials, written in English and available in full-text format, involving adult patients diagnosed with rosacea, regardless of clinical subtype.

Exclusion criteria included articles focusing on rosacea in the pediatric population, on patients with major comorbidities - particularly oncologic or autoimmune diseases - and studies exclusively evaluating dermocosmetic interventions in narrowly defined subpopulations such as those stratified by ethnicity or limited to specific geographic regions.

Results

A total of 652 publications were selected from PubMed but only 24 in accordance with the aim of this review. Thirteen were narrative or systematic reviews, 5 were clinical trials, 4 were guidelines, 1 was a prospective case-control study, and 1 was an epidemiological survey. A preliminary study investigating a topical serum was excluded from our review, as the product has not yet received FDA approval¹¹.

Discussion

According to our literature selection, four persistent gaps in the clinical management of rosacea were identified¹²:

1. Insufficient patient education regarding the role of daily skincare in restoring and preserving epidermal barrier integrity.
2. Challenges in selecting appropriate skincare products, often due to a lack of evidence-based guidance from healthcare professionals.
3. Limited patient awareness of the chronic nature of rosacea and the impact of lifestyle factors on disease progression.

4. The heavy psychosocial burden associated with the condition, that is frequently perceived by patients as socially limiting.

Rosacea is consistently described as a chronic inflammatory dermatosis, characterized by the sequential combination of periods of flare-up and periods of relative clinical quiescence. Flare-up phases may be triggered by several lifestyle-related factors¹³. During these periods the symptoms of rosacea become more noticeable. In the quiescent phase the symptoms improve, lessen or temporarily disappear, although a chronic inflammatory status persists with impaired skin barrier function and cutaneous microbiome dysregulation^{1,14,15}.

This dysfunction includes increased transepidermal water loss (TEWL), decreased stratum corneum hydration, elevated skin surface pH and increased sensitivity to lactic acid stinging tests. Inappropriate skin care behaviors, including both insufficient and excessive cleansing, have also been recognized as aggravating factors that may exacerbate clinical symptoms¹⁶. In light of these findings, the implementation of a structured and evidence-based skincare routine – encompassing the use of a gentle cleanser, a protective moisturizer and broad-spectrum photoprotection – is strongly recommended to support the skin microbiome, to maintain physiological pH and to reinforce the barrier function; dermatocosmetic formulations intended for use in rosacea should also ensure both efficacy and tolerability¹⁷⁻²⁰.

Our literature review exclusively focused on cosmetics that are defined as “articles intended to be rubbed, poured, sprinkled, or sprayed on, introduced into, or otherwise applied to the human body or any part thereof for cleansing, beautifying, promoting attractiveness, or altering the appearance” without affecting structure or function²¹⁻²³.

Furthermore, current dermatocosmetics may also include active substances²⁴. The same molecule could be a cosmetic constituent in one concentration, while it could be used as a cosmeceutical in another²⁵. The cosmeceuticals were deliberately not considered in our review.

Cleansers should be safe, with physiological pH (4.5-5.5), and devoid of additives, alcohol, fragrances,

perfumes or other sensitizing agents. In the flare-up phase the use of products requiring rinsing is not recommended. Therefore, a cleansing cream, milk, or lotion are the best choice. Cleansing should be performed gently and without rubbing or friction in order to minimize the risk of erythema and burning sensations. In the quiescent phase, for patients who like the sensation of rinsing, mild syndets may be used^{26,27}. Rinsing is recommended with lukewarm water to avoid thermal stress and additional irritation¹⁸. Cleansing brushes and scrubs, soaps with an alkaline pH are strongly discouraged as they can disrupt the epidermal barrier by depleting essential lipids, elevating skin surface pH and impairing barrier function.

Some natural mineral waters offer highly effective support both in the flare-up and in the quiescent phase. They may be applied by spraying after cleansing. Their effectiveness is related to the demonstrated relevant anti-inflammatory, immune-modulating and regenerative properties. Moreover, the cooling sensation is often well received by the patient²⁸⁻³⁵.

Creams constitute a key element in rosacea management, helping with the restoration of epidermal barrier integrity and improving stratum corneum hydration. The choice of formulations water in oil (W/O) vs oil in water (O/W) should be tailored to the patient's specific skin features.

In the flare-up phase a W/O soothing cream, aiming at protecting the skin from environmental stressors, is preferable in the morning, while a O/W moisturizing cream is recommended in the evening to improve skin hydration.

An O/W moisturizing cream is always recommended both in the morning and in the evening in the quiescent phase. Optimal formulations might include a combination of humectant and occlusive agents to replenish intercellular lipids and reduce TEWL. Ceramides, cholesterol, fatty acids, sphingolipids, borage oil, corn oil, and squalane-shea butter might be used as lipid-replenishing and emollient agents. Film-forming substances, such as hyaluronic acid, chitosans, peptides, and plant-derived mucilages, might be used as hydrophilic humectants.

Recommended creams may be enriched with anti-inflammatory and antioxidant agents, as well as active compounds such as vitamin B3-panthenol, vitamin E,

or products that promote and stabilize cutaneous microvascular function, such as rutin, vitamin K, anthocyanosides, ginkgo biloba, and saponins³⁶. Obviously, the use of products containing common sensitizers such as fragrances, preservatives and tocopherol should be strictly forbidden. Cell-proliferative agents, astringents, toners, camphor, menthol, alcohol or acetone are also contraindicated due to their potential to induce irritation and exacerbate barrier dysfunction.

In the quiescent phase the application of cream can be preceded, in the morning, by the use of an antioxidant serum containing vitamin C, vitamin B3, ferulic acid, ubiquinone, resveratrol, glutathione, astaxanthin, alpha-lipoic acid, proanthocyanidin. In the evening, a vascular defense serum is advised.

Photoprotection is an essential component of rosacea care, as ultraviolet radiation (UVA-UVB) is a recognized trigger of both inflammation and vascular dysregulation. Daily application of broad-spectrum sunscreens with a Sun Protection Factor (SPF) ≥ 30 is strongly advised. Inorganic filters such as zinc oxide and titanium dioxide have always been preferred to the chemical ones due to their superior tolerability in sensitive skin types. Moreover, according to FDA, only these sunscreens are recognized as Generally Recognized As Safe and Effective (GRASE)^{37,38}.

During quiescent phases, photoprotection helps mitigate reactive oxygen species (ROS)-mediated oxidative damage, while in active phases, it reduces the risk of photosensitization associated with systemic tetracyclines and topical agents like benzoyl peroxide¹⁹. Formulations should be fragrance-free, non-comedogenic and specifically designed for rosacea-prone skin. Tinted sunscreens may provide additional benefit by masking facial erythema and improving treatment adherence. The inclusion of anti-inflammatory and antioxidant ingredients – such as niacinamide, green tea extract or feverfew – might offer additional support to maintain skin barrier function. Application should be generous and uniform, and reapplication is recommended during prolonged or intense UV exposure.

Make-up can help immediately conceal skin imperfections and improve one's self-confidence. However, it should not aggravate the underlying dermatosis. Therefore, it should be avoided in the flare-up phase. Skin redness can be camouflaged with light green or

yellow coloured make-up. The base of these products should be adapted to the condition of the skin and must be free of mineral oils, fragrances and preservatives. Make-up should be applied at least 10 minutes after the administration of topical medications. High-coverage products containing fats and waxes are not recommended whereas talc-free mineral make-up provides excellent coverage and protects against UV radiation, and is non-comedogenic and anti-inflammatory, too. Mineral powders require the skin to be clean and prepared with a moisturizing cream, and thus it is necessary to wait until the cream is fully absorbed before their application¹⁰. As allergic reactions are possible with both natural and synthetic colorants, it is essential to perform a patch test before using new make-up products.

Education and behavioural measures are considered key components in a multidisciplinary approach to the management of rosacea, where effective communication between physicians and patients plays a crucial role¹⁶⁻¹⁹. The principle of 'knowledge to prevent' is central. Identifiable triggers such as sun exposure, hot beverages, alcohol, spicy foods, fatty foods containing cinnamaldehyde or rich in histamine, intense physical exercise, high-temperature environments and abrupt temperature changes, should be avoided. Additionally, improper use of dermocosmetic products, including harsh cleansers (e.g. surfactants, exfoliants), astringents and alkaline skin care products should be strictly avoided to prevent exacerbations. Furthermore, the accidental facial application of hair care products, such as shampoos, balms, hairsprays, foams, and hair dyes, must be carefully avoided.

Conclusion

Daily dermocosmetic care and appropriate lifestyle measures represent fundamental and complementary components to pharmacological treatment of rosacea.

According to the current literature, an appropriate up-to-date dermocosmetic protocol for rosacea is summarized in Table 1.

This regimen is applicable in both the acute and the remission phases. During acute exacerbations, the

Table 1. Dermocosmetic protocol in rosacea management^{15,24,29,31,36}.

Disease phase	Morning skincare	Evening skincare
Rosacea flare-up	Cream, milk or lotion cleanser	The same protocol
	Natural mineral water spray	The same protocol
	Topical medication: according to medical indication	Topical medication: according to medical indication
	<i>After 10 minutes</i>	<i>After 10 minutes</i>
	Soothing cream (W/O)	Moisturizing cream (O/W)
	<i>After 5 minutes</i>	
	Physical sunscreen SPF ≥30 and protective lip balm	
Rosacea in remission	Cream, milk or lotion cleanser or syndet followed by gentle lukewarm rinse	The same protocol
	Natural mineral water spray	The same protocol
	Antioxidant serum	Vascular defense serum
	Moisturizing cream (O/W)	Moisturizing cream (O/W)
	Physical sunscreen SPF ≥30 and protective lip balm	
	Corrective make-up	
To avoid	Prolonged exposure to heat sources (saunas, steam baths)	
	Sudden temperature changes	
	Peeling except for mandelic peel	
	Alcoholic or astringent products	
	Use of abrasive sponges	
	Rubbing the face vigorously	
Recommended	Broad-spectrum sunscreen daily SPF ≥30	
	Wide-brimmed hats	
	Sunglasses	
	Healthy lifestyle	
	Daily skincare	
	A diet rich in vitamins, antioxidant food and low in simple carbohydrates, alcohol, and food containing histamine and capsaicin as well as avoiding excessively hot beverages	

protocol aims to mitigate cutaneous inflammation and enhance tolerability to topical pharmacologic agents. In remission phases, it is intended to support maintenance therapy and to reduce the risk of relapse.

A successful combination of pharmacological therapies and cosmetic procedures requires a tailored, individualised approach, achieved through close co-operation between the Aesthetic Physician and the Dermatologist.

Conflict of Interest Disclosure: The authors declare that they have no conflict of interest.

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